

The basic principle of counting

When each stage is independent, multiply. That single sentence replaces a page of listing.

LESSON 86–88

3 × 40 MIN

ALGEBRA 8, CHAPTER IV §28

STAGE 9 · 12.2–12.3

LESSON CLOCK

6'

12'

8'

10'

4'

Recall yesterday's lists	6 min
The multiplication rule	12 min
Arrangements and factorials	8 min
Addition rule and "at least one"	10 min
Homework	4 min

LEARNING OBJECTIVES

- State and apply the multiplication principle of counting.
- Count arrangements of n different objects.
- Use the addition principle for cases that cannot happen together.
- Count by subtracting from the total when "at least one" appears.

TEXTBOOK

National	Chapter IV, pp. 189–197
Cambridge	Learner's Book pp. 261–270
Workbook	Workbook 12.2–12.3

01

Explanation

The multiplication rule

BASIC PRINCIPLE OF COUNTING

If a task is done in k stages, the first in n_1 ways, the second in n_2 ways and so on — **whatever was chosen before** — then the whole task can be done in

$$n_1 \cdot n_2 \cdot \dots \cdot n_k$$

ways.

Yesterday's café gave $3 \times 4 = 12$ meals, and the reason is now visible in the list: three equal blocks of four. The rule only replaces the listing — it does not change the answer.

THE CONDITION THAT MATTERS

The number of ways at each stage must not depend on *which* choice was made earlier. “Four shirts, three ties” multiplies. “Four shirts, and the number of matching ties depends on the shirt” does not.

Arrangements and the factorial

In how many orders can 5 different books stand on a shelf? Five choices for the first place, then four books remain, then three, two, one:

$$5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$$

This product has a name: $5! = 120$, read “five factorial”. In general

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 2 \cdot 1, \text{ and by agreement } 0! = 1$$

N	1	2	3	4	5	6	7
$n!$	1	2	6	24	120	720	5040

Factorials grow terrifyingly fast — $10! = 3\,628\,800$. That is exactly why counting by listing stops being practical and the rule takes over.

If only r of the n objects are placed, stop the product after r factors: seating 3 of 5 pupils gives $5 \cdot 4 \cdot 3 = 60$.

Adding, and counting the opposite

ADDITION RULE

If two cases **cannot happen together**, count each and add. Going to Samarkand by one of 3 trains **or** one of 5 buses gives $3 + 5 = 8$ ways — not 15, because you take one vehicle, not both.

Multiply for “and then”; **add** for “or”. Reading the word in the question is half the work.

“At least one” is almost always easier **backwards**. How many three-digit codes from 1, 2, 3 contain at least one 3?

$$\text{all codes } 3^3 = 27 - \text{codes with no 3 } 2^3 = 8 = 19$$

Counting the opposite and subtracting is called **complementary counting**. Use it whenever the direct count splits into many cases.

02 Worked examples

EXAMPLE 1 A car number plate has 2 letters (from 26) followed by 3 digits. How many plates are possible?

- ① 26 ways for the first letter
Letters may repeat unless the question forbids it.

- ② 26 ways for the second

- ③ $10 \cdot 10 \cdot 10$ for the digits
Three independent stages.

- ④ $26 \cdot 26 \cdot 10^3 = 676\,000$
Multiply every stage together.

ANSWER

676 000

EXAMPLE 2 In how many ways can 6 pupils stand in a queue, if two particular pupils must stand together?

- ① Treat the pair as one block
Now there are 5 items to arrange.

- ② $5! = 120$
Arrangements of the block and the other four.

- ③ $\times 2$
The two inside the block can swap.

- ④ $120 \cdot 2 = 240$

ANSWER

240 ways

EXAMPLE 3 How many four-digit numbers have all digits different and are odd?

① *last digit: 5 ways*

It must be one of 1, 3, 5, 7, 9 — deal with the strictest condition first.

② *first digit: 8 ways*

Not 0, and not the digit already used at the end.

③ *second digit: 8 ways*

Ten digits less the two used.

④ *third digit: 7 ways*

⑤ $5 \cdot 8 \cdot 8 \cdot 7 = 2240$

ANSWER

2240 numbers

03 Interactive model

Work through the counting tasks one stage at a time; ask the class for the next factor before revealing it.

Multiply, add, or subtract?

26 letters, 10 digits, repetition allowed. How many passwords?

① $26 \cdot 26 \cdot 26$

Three
letter
stages,
independent.

② $\times 10 \cdot 10$

Two
digit
stages.

③ $17\,576 \cdot 100$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Counting principle	Sanashning asosiy qoidasi	Основной принцип счёта
Multiplication rule	Ko'paytirish qoidasi	Правило умножения
Addition rule	Qo'shish qoidasi	Правило сложения
Independent choices	Bog'liq bo'lmagan tanlovlar	Независимые выборы
Factorial	Faktorial	Факториал
Permutation	O'rin almashtirish	Перестановка
Complementary counting	To'ldiruvchi hisoblash	Подсчёт через дополнение
Stage (step)	Bosqich	Этап
At least one	Kamida bitta	Хотя бы один
Total number of ways	Usullar soni	Число способов

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A shop has 5 jackets and 4 hats. How many jacket-and-hat outfits?

What is $6!$?

You may travel by one of 4 trains or one of 6 buses. How many ways?

How many three-digit codes from 1, 2, 3, 4 have no digit repeated?

64

24

12

48

How many of the $2^5 = 32$ outcomes of five coin tosses have at least one head?

31

16

32

30

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *Find $4!$*

ANSWER 24

2. *3 shirts, 5 ties — how many sets?*

ANSWER 15

3. *How many two-digit codes from 0–9, repeats allowed?*

ANSWER 100

4. *In how many orders can 3 books stand on a shelf?*

ANSWER 6

5. *2 roads from A to B and 3 from B to C. How many routes A to C?*

ANSWER 6

6. *A menu has 4 mains or 3 salads. Choose one dish — how many ways?*

ANSWER 7

7. *Find $5! \div 3!$*

ANSWER 20

Medium · the standard question

7 PROBLEMS

1. *How many three-digit numbers have all digits different?*

ANSWER $9 \cdot 9 \cdot 8 = 648$

2. *In how many ways can 5 pupils be seated in a row?*

ANSWER 120

3. *A code is 2 letters (26) then 4 digits. How many codes?*

ANSWER $26^2 \cdot 10^4 = 6\,760\,000$

4. *How many four-digit even numbers use digits 1–5 with repeats?*

ANSWER $5^3 \cdot 2 = 250$

5. *From 7 pupils, how many ways to choose a captain, then a vice-captain?*

ANSWER 42

6. *How many three-letter words from A–E have no repeated letter?*

ANSWER 60

7. *How many of the 36 two-dice outcomes contain at least one 6?*

ANSWER $36 - 25 = 11$

Hard · stretch and reason

7 PROBLEMS

1. *In how many ways can 5 people sit in a row if two of them must sit together?*

ANSWER $4! \cdot 2 = 48$

2. *In how many ways can 5 people sit in a row if two of them must NOT sit together?*

ANSWER $120 - 48 = 72$

3. *How many four-digit numbers with distinct digits are divisible by 5?*

ANSWER 952 — ending 0: $9 \cdot 8 \cdot 7 = 504$; ending 5: $8 \cdot 8 \cdot 7 = 448$

4. *How many three-digit numbers from 1–6 (repeats allowed) contain at least two 3s?*

ANSWER 16 — exactly two: $3 \cdot 5 = 15$, all three: 1

5. *In how many ways can the letters of MATH be arranged so that M comes first?*

ANSWER $3! = 6$

6. *A committee of a chair and 2 ordinary members is chosen from 6 people. How many ways?*

ANSWER $6 \cdot 10 = 60$ — chair 6 ways, then 2 from 5 gives 10

7. *Show that the number of subsets of a set with n elements is 2^n .*

ANSWER Each element is either in or out — n independent two-way choices, so 2^n by the multiplication rule.

07 Homework

Homework — 6 problems

Algebra 8, Chapter IV, pp. 189–197. Say which rule you used in each answer.

- Find $6!$ and $7!$
- A café has 3 soups, 5 mains and 2 desserts. How many three-course meals?
- How many three-digit numbers have all digits different?
- In how many orders can 4 photographs be hung in a row?
- How many three-digit codes from 1–5 contain at least one 1?

6. In one sentence, explain when you multiply and when you add.